Challenges in Automated Processing of Speech from Child Wearables: The Case of Voice Type Classifier

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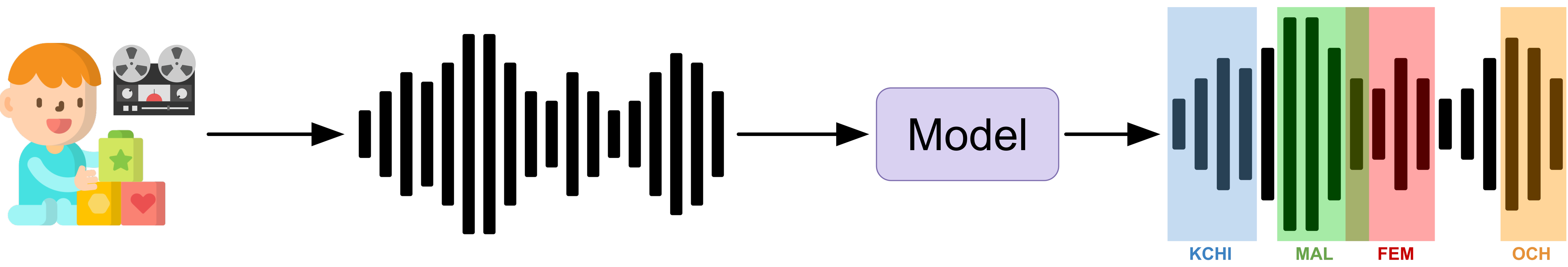
What is the Problem ?

Identifying who speaks when in child-centered long-form recordings.



***KCHI**: Key child, **OCH**: Other children, **MAL**: Adult Male, **FEM**: Adult Female*

Current State

	KCHI	OCH	MAL	FEM	Ave.
LENA (2004)	54.9	28.5	37.2	42.6	40.8
PyanNet-VTC (2020)	68.2	30.5	41.2	63.7	50.9
Human perfomance (inter-human agreement)	79.7	60.4	67.6	71.5	69.8

Table 1: F-scores obtained with automatic classifiers and human annotation. **Shows significant room for improvement**, especially for *OCH* and *MAL* classes.

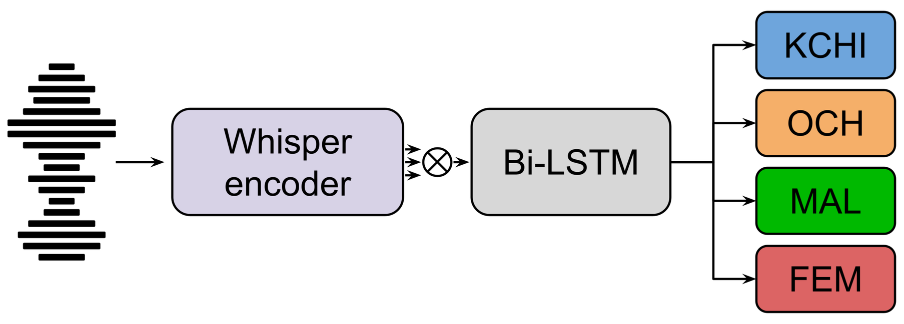
Our Approach

1. Dataset

Corpus	Tot. Dur.	KCHI	OCH	MAL	FEM
BabyTrain-2020	237h 4m	52h 45m	5h 27m	2h 22m	65h 52m
BabyTrain-2025 (ours)	669h 25m	158h 8m	11h 41m	12h 45m	262h 0m
Hold-out set	20h 0m	1h 39m	0h 45m	0h 43m	2h 48m

Table 2: Dataset size **increased** through **multi-institutional collaboration**.

2. Model



Whisper-VTC replaces SincNet with a pre-trained Whisper encoder, learning weighted features while leveraging robust multilingual representations.

What Worked

Effect of speaker category

	KCHI	OCH	MAL	FEM	Ave.
PyanNet-VTC (2020)	68.2	**30.5**	41.2	63.7	50.9
Whisper-VTC (tiny)	62,6	1.34	39.0	59.5	40.6
Whisper-VTC (base)	63.7	6.7	49.9	66.0	46.6
Whisper-VTC (small)	**68.4**	20.6	**56.7**	**68.9**	**53.6**

Table 3: F-scores (%) obtained on the hold-out, with varying Whisper sizes. Best performances are shown in bold, second bests are underlined.

Effect of data quantity

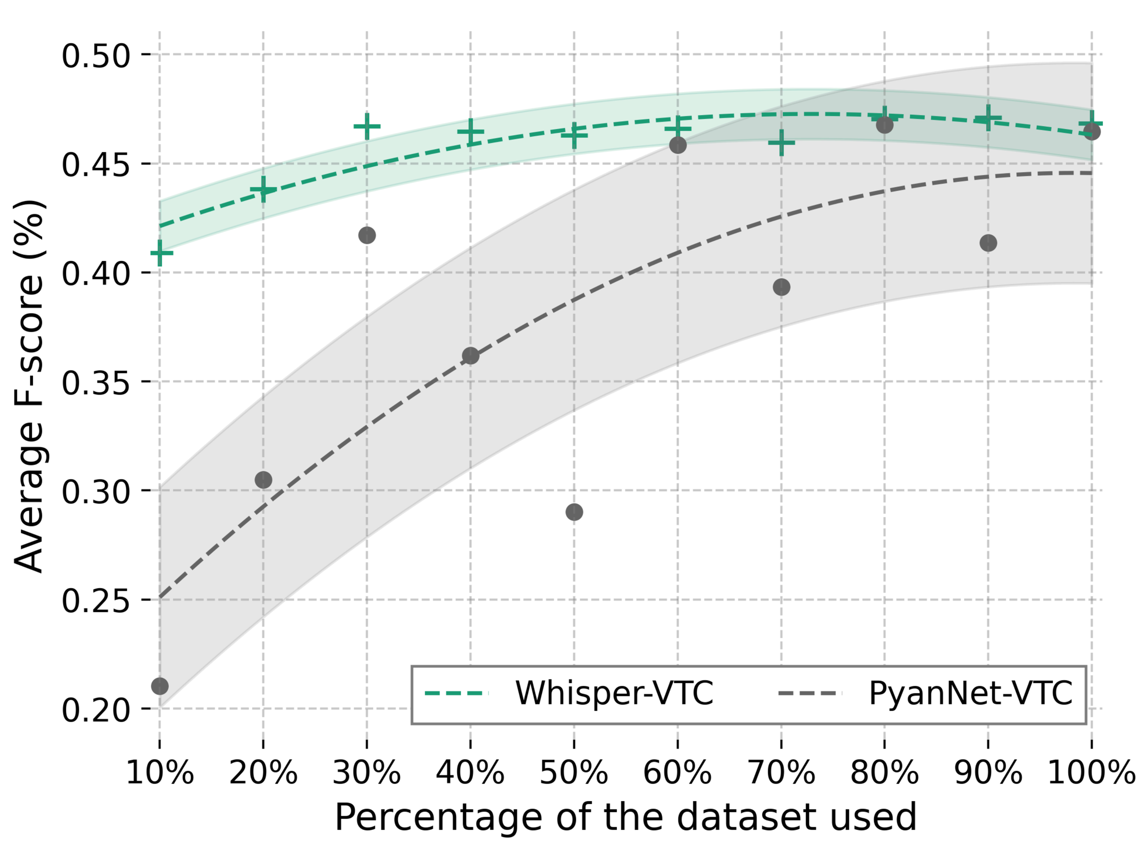


Figure 1: Whisper pretrained representations outperform training from scratch in low-data regime. **Overall, performance is much more stable.**

Effect of recording conditions

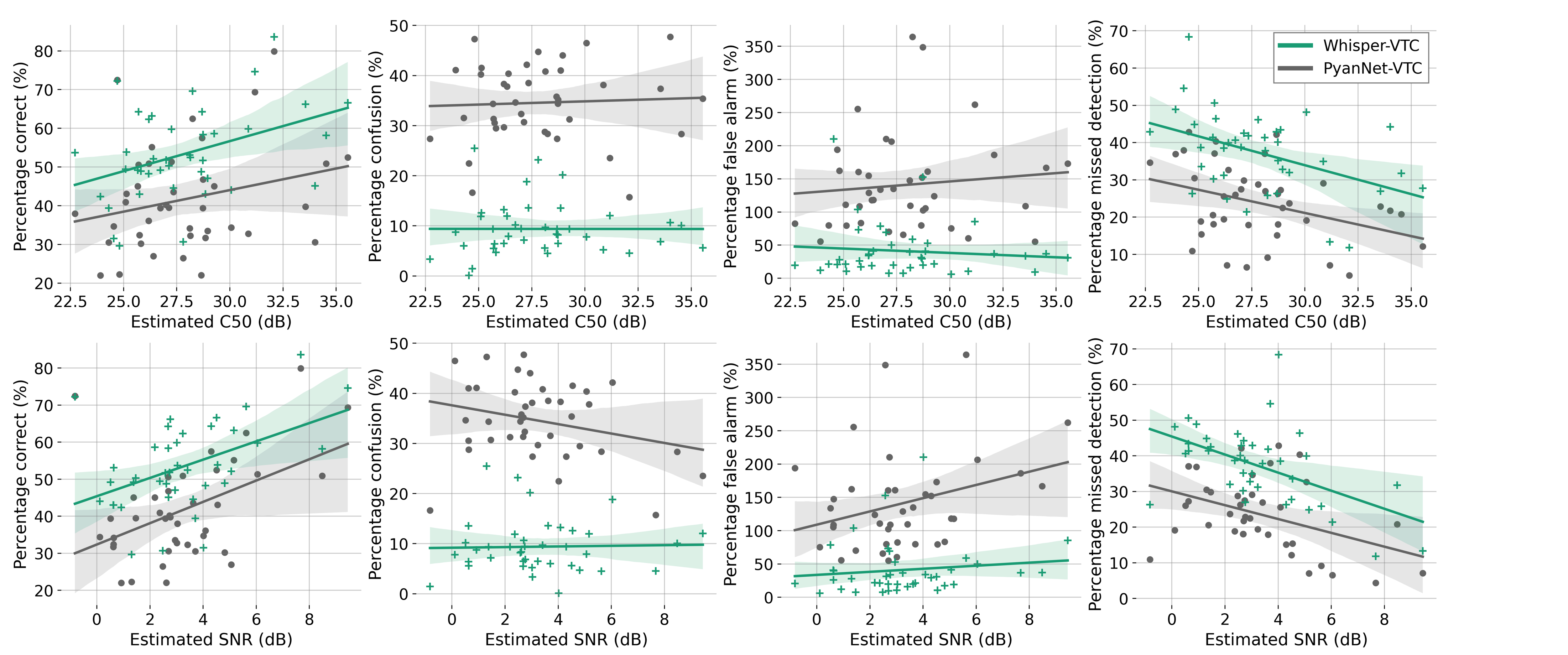


Figure 2: Miss (%), false alarm (%), confusion (%), and correct (%) obtained by **PyanNet-VTC** and **Whisper-VTC** (base) as a function of speech-to-noise ratio (SNR) and C50 (estimated by Brouhaha). Each point represents the performance on one child. Solid lines show linear regression with shaded 95% confidence intervals.

Our Past Experiments

Data augmentation techniques, underlying problem representation transformation, class imbalance techniques …

The "bitter lesson" applies: **sophisticated techniques have less impact than quality data and increased compute**.

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